Data visualisation report - League of Legends Season 13 data

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Abstract

This project has the goal to give insight into champion performance, popularity, and balance dynamics in the video game, League of Legends. It does so by using a publicly available dataset of ranked play statistics from season 13 (data for 2023). The study analyzes win rate, pick rate and ban rate across twelve patches to examine how individual champions, roles and classes can influence match outcomes over time. An interactive dashboard was developed using Plotly Express to visualize these metrics through bar charts, time-series plots, bubble charts, area charts, pie charts, and animated visualization. The analysis reveals that champions with high win rates are not necessarily the most picked or banned, suggesting that player perception and frustration plays a role in ban decisions alongside objective performance. Analysis of the classes also indicate dominance of Fighters and Marksmen in pick rates, while bans are disproportionately concentrated among a small subset of champions. Correlation between pick rate and win rate are weak, showing that popularity does not directly translate into how good a champion is. The project also evaluates the use of AI-tools in data visualization, finding that while they can automate graph creation, they currently lack interpretability and interactivity compared to manually designed solutions. Overall, the project demonstrates how data visualization can provide meaningful insights into this complex game.

Background and motivation

The group decided to go with a dataset from a popular video game that both group members have invested a significant amount of time playing. While the decision to go with statistics from a video game might seem as an odd choice, there is quite a substantial amount of quality data available, and so the decision was made to stick to the League of Legends Season 13 champion statistics dataset.

League of Legends[1] is a five versus five team based strategy game, where the goal is to destroy the opponents “nexus”. This is done by choosing a playable character, a “champion”, where players must work together to destroy structures and defeat other champions in order to make their way into the enemy base and take down the nexus. Each person has a specialised role and position on the playable area, the “map”. Before each game, each player “ban” a champion from the game that is being set up, before choosing their own champions. This means that a banned champion cannot by any player in the incoming game. This is a strategy option and is typically used to remove problematic champions from the game that is about to start.
As the game is constantly evolving, each year of play, a “season” consists of monthly updates, “patches”, that introduce changes to some of the champions in order to change the amount of agency of the game they have, by changing certain attributes.
One important aspect is that this data is from Season 13, which took place in 2023. As of writing, the current League of Legends season is Season 15. Despite this time discrepancy, the data still has potential to give valuable insight, as “Meta”’s,. Most Efficient Tactic Available, doesn’t always change in significant ways and stays relatively consistent.

As the game is incredibly complex and has a huge amount of variables, even the smallest change to a champion affects them in significant ways, which also affects if players choose to use them, “ban” them which removes them from potential play in a game, or ignore them and pick other champions instead. These choices significantly affect the ending outcomes of a game, and as the game is a hobby that the group participates in quite frequently, it would help make informed decisions on what champions should be played, which should be banned, which should be avoided and which can be ignored as their agency is negligible. All in all, working with data would aid in making informed decisions in future play for the group, and hopefully help in winning more games, as it is clear that the group could benefit from that kind of information.

Project Objectives

The main objective of this project is to analyze and visualize how champion performance and popularity changes through time. By analyzing statistics the project aims to uncover trends in champion effectiveness to win a game, role dominance and meta shifts over time. To uncover the project goals these primary questions have been made to deduct results from the dataset:

1. Which champions dominate the meta in terms of performance and popularity within a patch.
    a. Which champions achieve the highest win rates in a given patch?
    b. Which champions are most frequently picked by players?
    c. Which champions are most commonly banned?
2. How do champion performance and popularity evolve across patches?
    a. How do individual champions' win rate, pick rate and ban rate change over successive patches?
    b. Do balance updates lead to gradual trends or shifts in champion performance?
3. Does popular champions have a positive effect on the win probability of the game?
    a. Is there a correlation between pick rate and win rate?
    b. Are certain champions or roles heavily picked despite having low win rates?
    c. Do some classes or roles achieve strong win rates without high pick rates?
4. How does champion class influence performance, popularity and bans?
    a. Which champion classes perform best in terms of win rate?
    b. Which classes dominate the meta based on pick rate share over time?
    c. Which classes attract the highest ban rates?
5. How is champion strength distributed across tiers and roles?
    a. Which champion classes contain most high tier champions?
    b. How evenly are bans distributed among champions?
    c. Are bans concentrated among a small set of champions or spread across the roster?
6. Can AI-generated visualization provide meaningful insights compared to traditional methods?

The objectives of this project is designed to gain insight in champion performance, popularity and balance dynamics in League of Legends across multiple patches. Each question will be answered with a specific visualization in a way that is measurable and visually interpretable. Together these objectives create a clear guide on how to develop the dashboard for examining the evolving state of the game.

Data

The dataset is a publicly available dataset from Kaggle[2]
https://www.kaggle.com/datasets/vivovinco/league-of-legends-stats-s13

The dataset contains champion statistics from the Season 13 Ranked play of League of Legends. Ranked play is the competitive environment, where players perform to the best of their ability and taking the game as serious as possible, meaning that game modes that might skew these stats have been excluded, in order to provide accurate data for the most popular mode of play

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The data is comprised 12 .csv files with 11 columns with 100+ rows, which includes:

- Name : Name of the champion
- Class : Fighter, Assassin, Mage, Marksman, Support or Tank
    - Classes define the playstyle of champions
- Role : Top, Mid, ADC, Support or Jungle
    - Roles define the objective and location of the player, when choosing champions that fit into this role
- Tier : God, S, A, B, C or D
    - A metric for briefly describing the efficiency of the champion
- Score : Overall score of the champion
- Trend : Trend of the score
- Win % : Win rate of the champion
    - Average winrate of the champion across the entire patch, these values are usually quite close to 50% as you either win or lose with a champion. Rates <49% and 51%> are seen as the champion having a significant effect on the outcome of the game.
- Role % : Role rate played with the champion
    - Describes the rate of which roles play a champion, this is due to some champions being viable in more than 1 role. For example, Amumu, A tank, can be played as both a Jungler, and Support. So their role %'s could be split 49% as support and 51% as jungler, meaning that players play the champion slightly more as a jungler than support, but still both are viable.
- Pick % : Pick rate of the champion
    - % of games the champion is picked
- Ban % : Ban rate of the champion
    - % of games the champion is banned and excluded from play in an oncoming match.
- KDA : (Kill+Death)/Assist ratio of the champion

The datasets are mostly not in need of any processing or cleanup. However, a python script was used to generate a separate .csv file with the combined data from all patches, with an added column denoting the patch the data is derived from to be used as an indicator of time.

Visualization/Dashboard

Design

The dashboard is designed with the goal to describe that data in a way that makes it clear which champions perform exceptionally, champion popularity and in general how the meta evolves with each patch. The design itself will adopt a multi-page layout to organize the different visualizations into distinct aspects of the analysis.

This should be done by taking the data and displaying it in a way that is easily digestible, where users are able to deduct relevant information. As the data encompasses a huge amount of different champions, over 140 individual champions with some repeating as their stats for different roles are different, a singular visualisation would not be able to communicate the diffePlotly Express[3] as the main framework, making use of their API.

To solve this issue, we will make use of a multi-page dashboard, where each tab displays a different type of data visualisation. These tabs should contain one of more visualisations of the same type, such as pie charts, bubble charts, graphs etc. There should be visualisations focusing on which champions are leading in terms of winrate, pickrate, banrate for a given patch, as well as a larger overview of all champions statistics relative to each other. It should be possible to choose between which patch the data is being derived from, and in some cases it should combine the data from all the patches to give a more comprehensive look. This will reduce any visual clutter of having different types of graphs close to each other, and allow potential users to more easily explore the data in different ways, while still keeping a clear and obvious purpose for the visualisation. As the winrate metric between champions is also relatively tight, it is important that this is being accounted for in the visualisations, displaying a clear difference between 45-55% while not overcompensating making the visualisations appear untrustworthy.

Must-Have Features

The following features were identified as necessary for the project to be a success. The absences of any of these features would result in an unfinished project.

The Dashboard must include at least three different types of graphs, to make sure that different perspectives are supported. Using multiple chart types allows the project to visualize metrics like ranking, correlation, composition or temporal change in the data. Furthermore, an animated graph is included to visualize movement in data, like how champion win rates evolve across patches. An AI-generated visualization will also be incorporated into the dashboard. This visualization will be produced using an external AI-tool and through a structured prompt. The purpose of including an AI-generated graph is first, to explore whether automated visualization tools can meaningfully represent relationships within the dataset, and second, to compare AI-assisted outputs with manually designed visualizations.

The final dashboard must also contain at least nine graphs in total to ensure that the project sufficiently covers the dataset and explore different metrics. A direct link to the deployed dashboard must also be included in the final report for easier accessibility. And at last the dashboard must provide an option to download the report as a manual, allowing the user to access the analysis of the data.

These must-have features establish a clear set of functional and analytical requirements that guide the design and implementation of the project.

Results

In order to display which champions dominate the meta within terms of performance and popularity within a patch, we implemented three simple bar grapes as seen below in figure 1

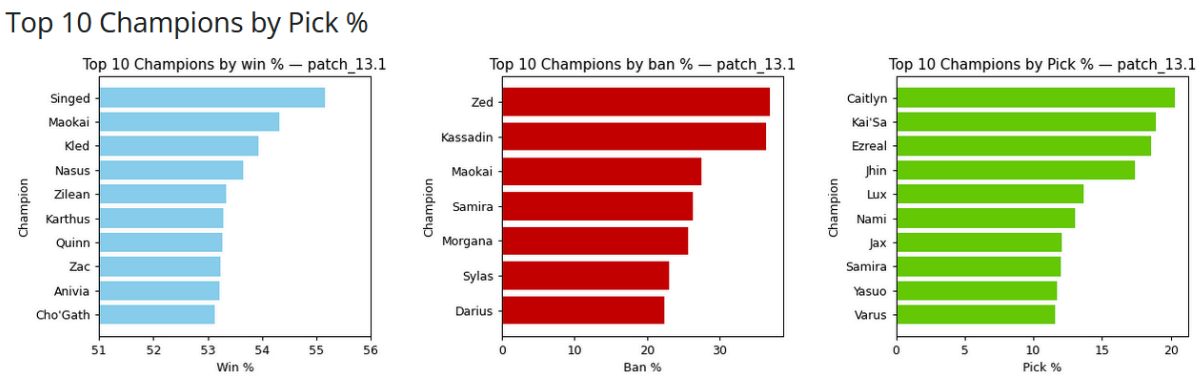

Figure 1 - Bar graph

This clearly provides a clear picture of the top 10 champions in terms of win rate, ban rate and pick rate. Changing the patch at the top of the page updates these graphs to correspond with the chosen patch, as the top 10 can change significantly with every patch. By visualising the data in this way, we see that there isn't always a clear correlation between champions that are picked and banned often, compared to their winrate. In some patches, this indicates that the champions with the highest win rate aren't always the most popular. It also displays that overall, players ban champions that might be perceived as incredibly strong or potentially frustrating to play against and not necessarily based on objective performance.

To find out how individual champion performance evolves across patches, A time series plot was implemented. The time series plot consists of different parts: A role selector to narrow down the champions that are relevant for a specific role, this was done to make easier to find relevant champions for the role. A list of champions that can be chosen to view, toggle buttons to show the win %, pick % and ban %, for easier filtering information the user deems relevant. The plot itself, which displays the chosen information, as seen in Figure 2.
One weakness of this type visualisation is that it is not currently possible to compare with more champions, a future integration of this feature could provide even more detailed insight.

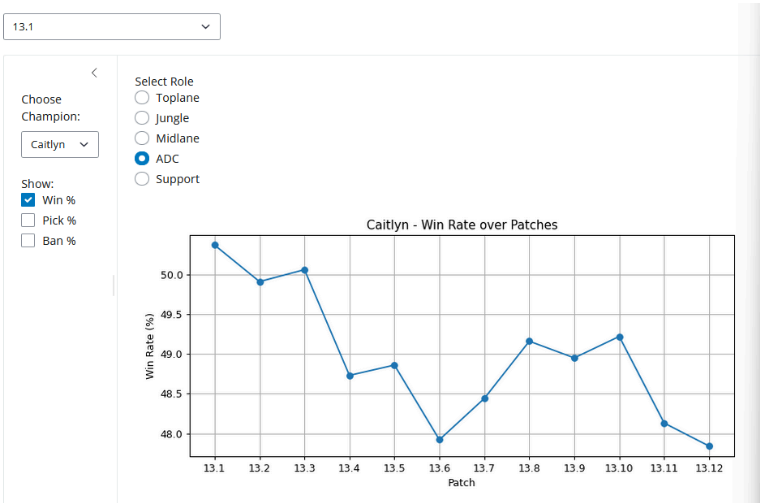

Figure 2 - Time series plot

For us to answer if popular champions have a positive effect on the win probability of games, three bubble charts were implemented to provide information in different ways. It was also decided that these charts would use the data for all champions to have a more comprehensive view that provides the full picture.

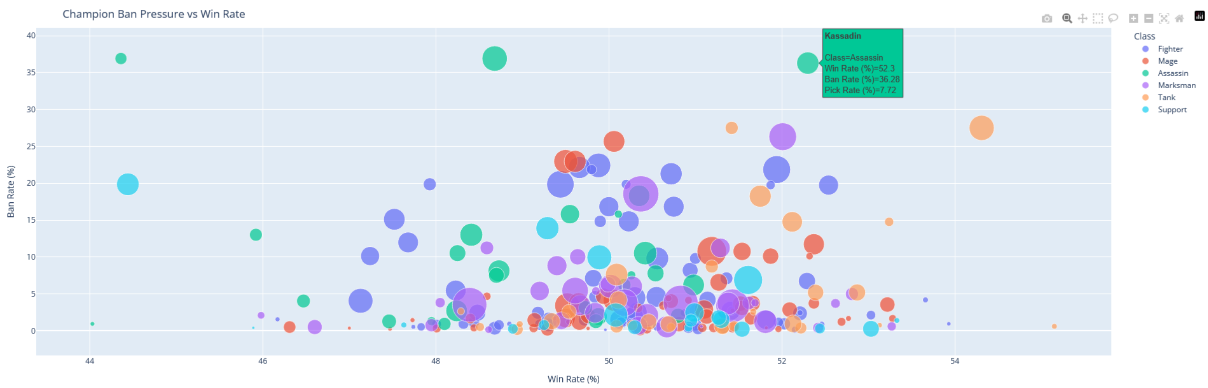

Figure 3 - One of the three bubble charts

Figure 3 displays one of the three bubble graphs, each bubble corresponds to a different champion, the color indicates the class, x-axis indicates winrate and y-axis indicates ban %. Each bubble contains more detailed information about the champion, where size of the bubble represents the pick % of the champion. As can be seen, the highest and lowest winrate champions aren't always banned, but not always popular choices for players either. While we are generally satisfied with the different bubblecharts, the second does have some weakness in terms of clearly displaying which roles have better winrate. This is due to there always being one of each role present in a game, meaning that differences in win rate is difficult to display, and could indicate that the role itself is not necessarily a huge contributing factor in winning a game, in the same way that champion picks are.

In order to display how champion class influences performance, popularity and bans, as well as how champion strength is distributed across tiers and roles, a mixture of area charts and pie charts was implemented.

Firstly, an area chart (figure 4) was implemented to display which classes are more often picked every patch. While it shows a clear preference for fighters and marksman champions, there isn't a huge difference between patches, which is most likely due to classes being closely tied to the role. This means that while the chart fulfils its purpose, the information that can be derived from it might not be all too useful for the user.

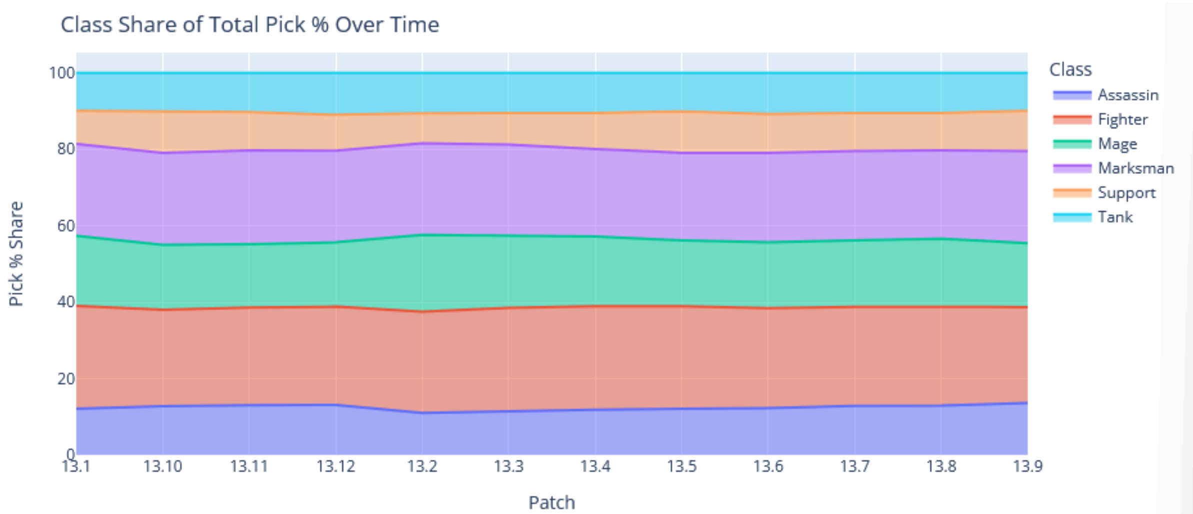

Figure 4 - Area chart of class popularity across patches

Three pie charts implemented. They display which classes are banned the most and the least in every patch, the distribution of God-tier champions inhabit which classes and the ban distribution for every champion, grouping champions below 1% ban rate into "Other champions" and clearly distinguishing that slice from the rest. The patch is controlled from the top of the page in the "choose game patch" list, similar to the bar graphs and bubble graphs.
From the Ban % share by champion class pie chart, figure 5, it clearly displays which classes are the most and least banned. For example for patch 13.1, a fighter or assassins is banned in over half of the games for the season, with support champions not even taking up 5%. Comparing this information with the Tier-share pie chart in Figure 6, this also correlates with fighters taking up a significant rate of bans, but assassins being less prevalent in God-tier compared to how often they are banned, at least for the patch 13.1. Using these charts in combination, users are able to make an informed decision on which Classes should be focused on bans more or less, and how that changes between every patch.

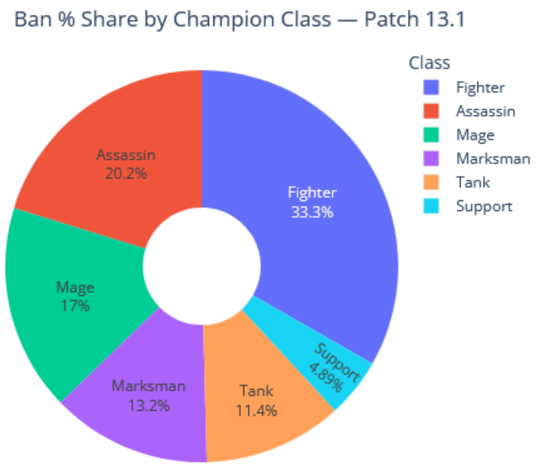

Figure 5 - Ban % share pie chart

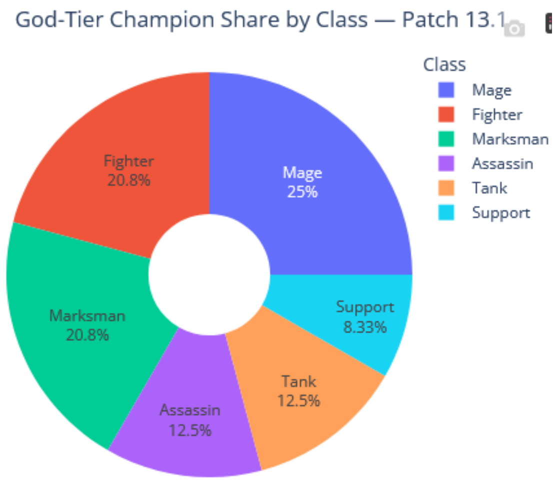

Figure 6 - God tier champion share chart

The final pie chart displays which specific champions are banned the most for the patch. As there are over 140 champions, any champion with a ban rate of less than 1% is grouped into the "Other Champions" slice. This is ideally used in combination with the two first ones, to better understand what specific champions could be chosen when wanting to play a game, In order to pick a champion that is very strong and isn't banned too often, to ensure that can actually play the champion that has a higher probability of winning. For patch 13.1, one thing that can be noticed is that the two most banned champions are Zed and Yasuo, respectively an assassin and a fighter. This displays some correlation between ban and strength, although this should be considered in combination with the previous charts.

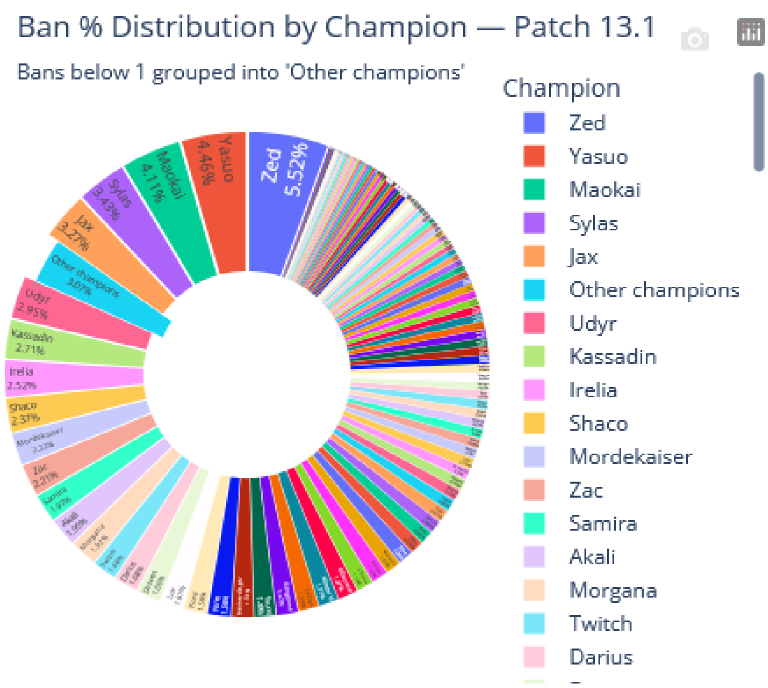

Figure 7 - Ban % dist. Chart

An AI generated chart was also attempted. ChatGPT 5.0 was provided with patch_13.1.csv and given the prompt:

> "I want you to generate a graph based on the csv file I just provided. It should display how likely the champion is to be banned, the higher the win%"

This resulted in the graph seen in figure 8.

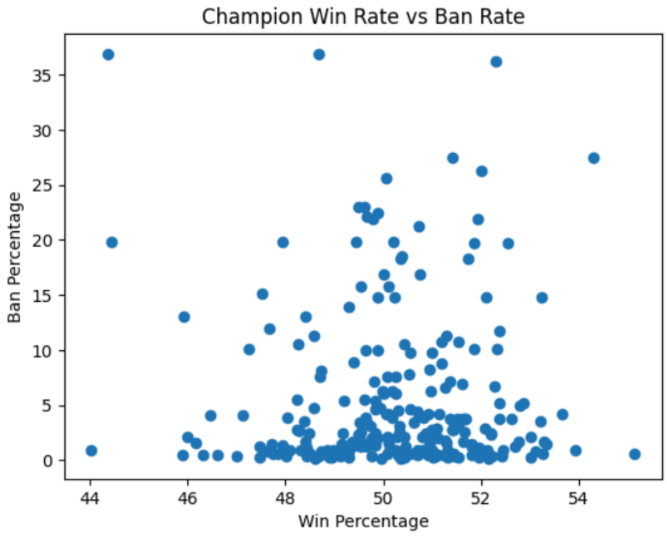

Figure 8 - AI generated plot

An obvious issue is that we are not able to see which champions are represented, and because it is an image, we are not able to hover the different dots for more information. Due to the limitations of a free plan of ChatGPT, it is not possible to provide more than 10 files, meaning that attempting to get a picture of the entire season would provide incomplete data. However, even limited to a single file, the data is not decipherable.

Finally, an animated bubble graph was implemented. This displays every champion represented as a bubble, which is colored to match its class, its size determined by ban %, x-axis position determines pick % and y-axis determines winrate. All the .csv files for all the 12 patches were combined into a single file, with an extra column added, which denotes the patch number as a measure of time. This was done to enable the ability to animate the plot, as Plotly requires some kind of measure of time. While the plot does display the information that we wanted, it has a couple of issues. There are only 12 points of time, meaning that any gradual change between patches is not observable, and the datapoints move around frantically, making it relatively hard to keep track of a single champion over the year. Another issue is the perceived smoothness of the animation and how it is implemented. Running on a local machine, clicking the "Animated" tab opens both the tab, and a separate window where the animation can be run as smooth as possible. Using a python script, the graph has been embedded in an iframe on the page. This is to make it possible to view the

animation from where the project has been hosted, Shinyapps[4]. The embedded solution can appear uneven depending on the browser used to view it.
To solve this issue, a fundamental change of the architecture would have to take place, in order to smoothly display an animated graph without any of the aforementioned "hacky solutions".

The entire project can be viewed from this link:
https://goodstudent.shinyapps.io/datavis-leagueoflegends/

Conclusion

This project focused on creating a dashboard for displaying League of Legends Season 13 ranked champion data, to allow users to view easily understandable data to gain insightful information about what champions dominate certain patches, as well as potentially help users make well informed choices about what champions to play. The dashboard developed using Plotly Express and contains Bar charts for top performing, banned and picked champions, time series plot for champion evolution across patches. Bubble charts for displaying the relationship between win rate, pick rate and ban rate. Area and pie charts for visualising class dominance, bans and tier distribution. Finally, an AI-generated visualization and an animated visualisation to show champion metric changes across all patches.
Users are able to filter by champion, patch, role, and different metrics.
Some challenges were encountered, such as tight data ranges in regards to winrate, the AI-generated visualisation having a lack of interactivity and poor interpretability, and some technical limitations of the chosen technologies when trying to implement animated visualisations in Plotly Express and hosting on shinyapps.
This project demonstrated even hobbies can include incredibly complex information and statistics, the visualisation of which can provide meaningful insight to those partaking in the activity. While some aspects of the project in its entirety can be improved with more development in the future, and perhaps working with datasets being generated from games in real time would provide more up to date insight.

References

[1] https://www.leagueoflegends.com/en-gb/how-to-play/

[2[https://www.kaggle.com/datasets/vivovinco/league-of-legends-stats-s13

[3] https://plotly.com/python/plotly-express/

[4] https://www.shinyapps.io/